Hydroelectric Power Plant Digital Twin Project

Technical Project Report Referencing Kavşak Bendi Hydropower Plant

Project	**Hydroelectric Power Plant Digital Twin**
Reference plant	**Kavşak Bendi HPP – Enerjisa Üretim**
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Abstract

This project presents an educational and portfolio-oriented digital-twin model of a hydropower plant, using Kavşak Bendi HPP as its reference. It examines the main energy-conversion process, operating behaviour, monitoring and fault scenarios, and grid connection. Python-based calculations in Google Colab are combined with a MATLAB/Simulink modelling approach. Scenarios are built around flow rate, net head, turbine-generator efficiencies, generation profiles, reservoir level and electrical connection parameters.

1. Aim and scope

The objective is not to reproduce every detail of the actual plant. Instead, the project represents the main physical relationships between plant components in a transparent and calculable model. The scope covers conversion of water energy into electricity through turbines and generators, unit coordination, daily generation behaviour, measurement and alert logic, and power transfer to the grid through a step-up transformer.

2. Reference plant and model inputs

The reference configuration consists of three main Francis turbines and one ecological-flow turbine. The main units are rated at 61.95 MW each, the ecological unit at 5.43 MW, and the total installed capacity is taken as 191.28 MW. Some calculation scenarios use a gross head of 100 m, a hydraulic loss of 5 m and a net head of 95 m. Efficiencies, flow limits and daily operating profiles are illustrative model inputs, not verified operating records.

3. Method and governing equations

Hydraulic power is calculated as $P_h = \rho g Q H_{net}$, where ρ is water density (approximately 1000 kg/m³), g is gravitational acceleration (9.81 m/s²), Q is flow rate (m³/s) and H_{net} is net head (m). Approximate electrical output is $P_e = \rho g Q H_{net} \eta_t \eta_g$, where η_t and η_g are turbine and generator efficiencies. For a three-phase system, apparent power is $S = \sqrt{3} V I$ and power factor is $PF = P/S$. Approximate line active-power loss is $P_{loss} = 3 I^2 R$, while approximate voltage drop is $\Delta V \approx \sqrt{3} I(R \cos\varphi + X \sin\varphi)$.

4. Project stages

4.1 Project setup and plant parameters

The Colab environment was prepared using libraries such as NumPy, Pandas and Matplotlib. Core reference parameters for Kavşak Bendi HPP—including installed capacity, turbine arrangement, head and efficiencies—were defined. Parameters not supported by operating records were treated as model assumptions.

4.2 Hydraulic and electrical power calculation

The relationship between water power, turbine output and generator output was established using flow rate and net head. The energy-conversion chain and the effect of efficiency on electrical output were examined.

4.3 Turbine modelling

The three main Francis turbines and one ecological-flow turbine were represented as separate units. Unit ratings and total installed capacity were considered, using 191.28 MW as the reference plant capacity.

4.4 Flow, head and efficiency analysis

The effect of changing flow rate and net head on available generation was calculated. Turbine and generator efficiencies were included to estimate the conversion from hydraulic power to electrical power.

4.5 Performance curves and capacity limits

Power curves were produced for different flow conditions. Capacity constraints near nominal power and flow limits were considered.

4.6 24-hour operating simulation

An hourly generation profile was modelled for changing daily flow/load conditions. The power profile and daily energy total were used to examine time-dependent plant behaviour.

4.7 Reservoir and water-level effects

The relationship between reservoir level and available head was considered. The effect of water-level changes on net head and generation was modelled.

4.8 Sensors and real-time monitoring concept

Flow, water level, pressure, power and related variables were treated as measurable digital-twin signals. The monitoring approach was examined using example or simulated measurements.

4.9 Fault and anomaly scenarios

Normal operation was compared with abnormal conditions. Threshold-based alerts were used to illustrate how unusual flow, pressure or power conditions could be detected. This is an educational logic, not a real protection setting.

4.10 Energy generation and operating optimization

Load/flow allocation between units and operating conditions were evaluated. The objective was to demonstrate generation improvement within defined limits; efficiencies and limits are model inputs.

4.11 MATLAB/Simulink modelling and comparison

Representation of the energy-conversion chain in MATLAB/Simulink and comparison with calculation-based modelling were considered. Block representation, signal scaling and unit consistency were key concerns.

4.12 Step-up transformer and grid connection

The 13.8/154 kV step-up transformer was examined in terms of voltage ratio, three-phase currents, active/reactive power, losses and efficiency. Current, line losses and approximate voltage drop were calculated for a 154 kV connection. Overcurrent, differential and earth-fault protection concepts and breaker responses were evaluated using example scenarios.

5. Transformer and grid connection – interpretation of example results

The final stage examined voltage transformation from 13.8 kV to 154 kV, giving a ratio of approximately 11.16. At rated voltage, the three-phase current is approximately 3,347 A on the low-voltage side and 300 A on the high-voltage side for 80 MVA. For 100 MVA, the corresponding values are approximately 4,184 A and 375 A. These are calculations based on nominal ratings and idealized voltage assumptions.

In an illustrative loss case, iron loss was set to 80 kW and copper loss at 75% loading to 225 kW, resulting in 305 kW total transformer loss. In a separate line example, assumptions of 154 kV, 20 km length, 0.10 Ω/km phase resistance and 0.90 power factor produce line losses on the order of 0.54 MW. The voltage-drop calculation is an approximation for a balanced three-phase system. These are scenario results, not field-test measurements.

6. Assumptions, limitations and validation needs

Key limitations include the absence of real-time SCADA data, manufacturer performance curves, verified turbine characteristics, waterway transient parameters, actual line impedance and relay coordination settings. Numerical results should therefore be treated as educational simulations and preliminary engineering estimates. Fault thresholds are illustrative and must not be used as actual protection settings. Different loading scenarios were used in some Stage 12 calculations; loss and efficiency values must be reconciled at one common operating point before being combined in a final integrated power-flow result.

7. Conclusion

The project systematically considers the main hydropower energy chain, from hydraulic input to power delivered to the grid. It examines the effects of flow and head, multiple turbine units, daily generation profiles, reservoir level, monitoring and fault logic, and transformer/grid connection within one modelling framework. The key learning is that hydropower operation cannot be understood through turbine output alone; hydraulic, mechanical, electrical and protection systems must be assessed together.